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remains HIGH for a time T_2 (pulse width of IC2) and then goes LOW (↓) which triggers IC1. This process is continuous and a square wave is produced at Q_1 and Q_2. The frequency of the square wave can be controlled by the timing elements.

9.6 555 TIMER

IC 555 timer (available in 8 pin DIP or TO-99 package) is one of the most popular and versatile sequential logic devices which can be used in monostable and astable modes. Its inputs and output are directly compatible with both TTL and CMOS logic circuits. The functional block diagram of 555 timer is shown in Fig. 9.33.

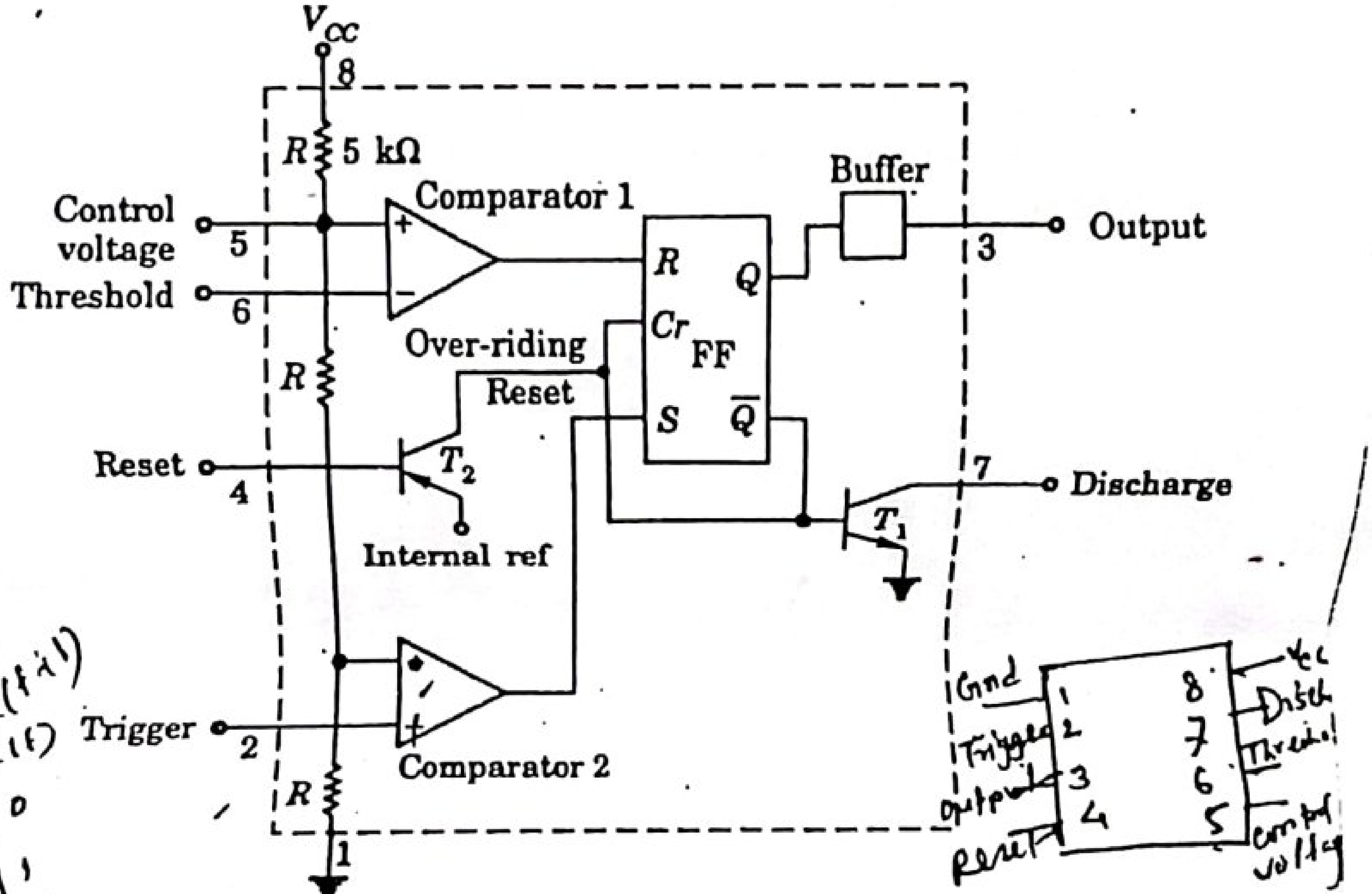

Fig. 9.33 Functional block diagram of 555 timer.

It has two comparators which receive their reference voltages by a set of three resistances connected between the supply voltage V_{CC} and ground. The reference voltage for comparator 1 is $2V_{CC}/3$, and for comparator 2 is $V_{CC}/3$. These reference voltages have control over the timing which can be varied electronically, if required, by applying a voltage to the control-voltage input terminal. If this is not required, then a bypass capacitor (0.01 μF) should be connected between this terminal and ground to bypass noise and/or ripple voltages from the power supply.

On a negative-going excursion of the trigger input, when the trigger input passes through the reference voltage $V_{CC}/3$, the output of the comparator 2 goes HIGH and sets the FLIP-FLOP ($Q = 1$). On a positive-going excursion of the threshold input, the output of the comparator 1 goes HIGH when the threshold voltage passes through the reference voltage $2V_{CC}/3$. This resets the FLIP-FLOP ($\overline{Q} = 1$).

The FLIP-FLOP is cleared (irrespective of the S input) when the reset input is less than about 0.4 V. When this input is not required to be used, it is normally returned to V_{CC}.

An external timing capacitor, C is to be connected between the discharge terminal and ground. When the FLIP-FLOP is in the reset state, its $\overline{Q} = 1$, which drives T_1 to saturation thereby discharging the timing capacitor. The timing cycle starts when the FLIP-FLOP goes to set state and therefore T_1 is OFF. The timing capacitor charges with the time constant $\tau = R_A \cdot C$, where C is the timing capacitor and R_A is an external resistor to be connected between the discharge terminal and V_{CC}.

The output is at logic 1 whenever the transistor T_1 is OFF and at logic 0 when T_1 is ON. The load can be connected either between the output terminal and V_{CC} or between the output and ground terminals. The four possible output connections are shown in Fig. 9.34. The maximum sink or source current is 200 mA. The voltage corresponding to HIGH output is approximately 0.5 V below V_{CC} and for LOW it is approximately 0.1 V. IC 556 timer contains two 555 timer circuits and is available in 14 pin DIP.

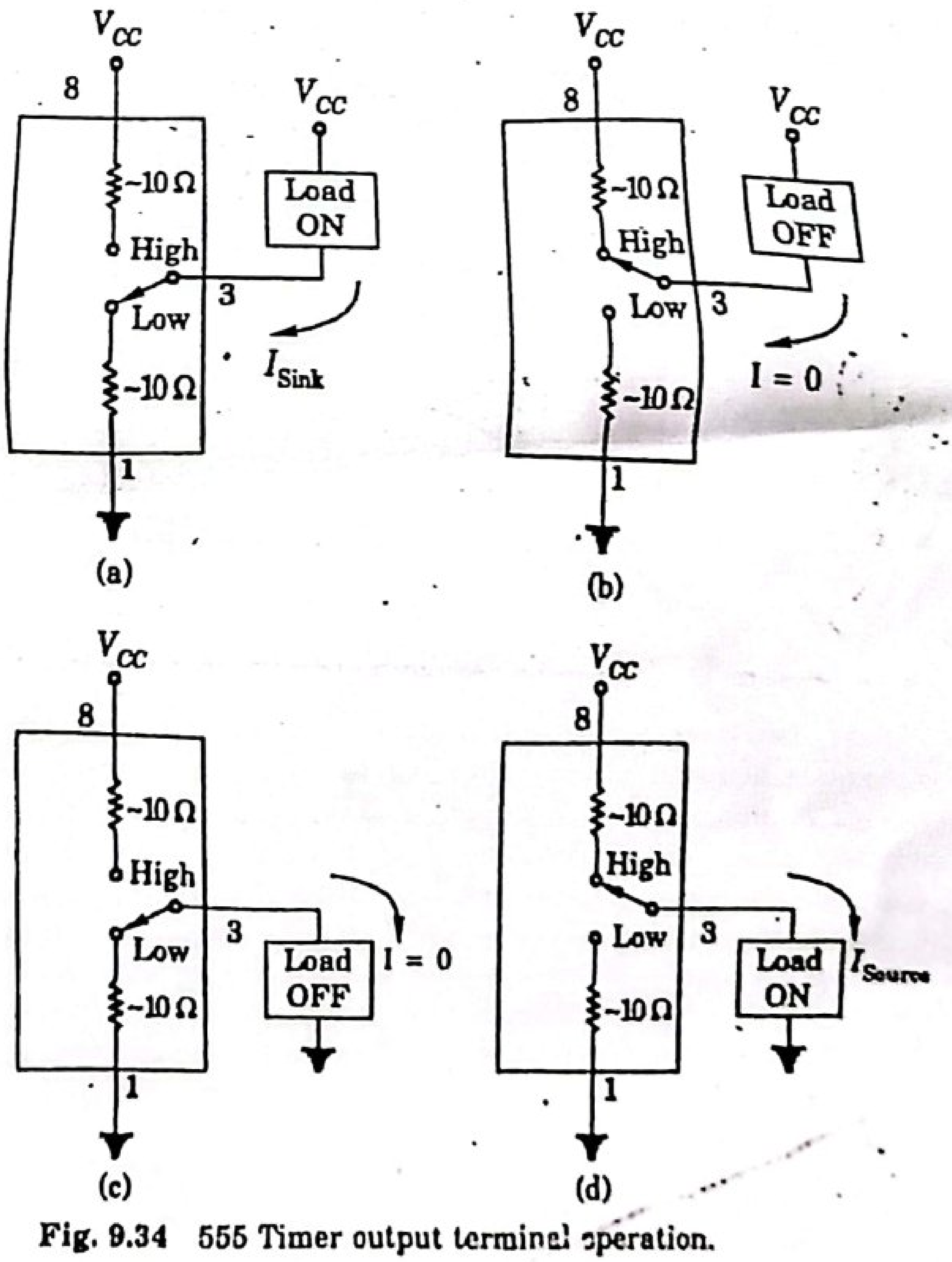

Fig. 9.34 555 Timer output terminal operation.